1. $ext(A) = int(\mathbb{R} \setminus A)$

$int(A) \cap ext(A) = \emptyset \quad \wedge \quad int(A) \cap fr(A) = \emptyset \quad \wedge \quad ext(A) \cap fr(A) = \emptyset$

$int(A) \cup ext(A) \cup fr(A) = \mathbb{R}$

$\bar{A} = int(A) \cup fr(A)$ (pontos aderentes ou fecho)

A' → intervalos de números do conjunto A (pontos de acumulação ou derivado)

$isol(A)$ → pontos soltos do conjunto A.

A é aberto sse $A = int(A)$.

A é fechado sse $\bar{A} = A$.

$Maj(A) \Rightarrow$ Existe $M \geq A$

$Min(A) \Rightarrow$ Existe $m \leq A$

A é limitado se $Maj(A) \wedge Min(A) \neq \emptyset$

2. Progressão Aritmética : $x_n = x_1 + (n-1).r$

Progressão Geométrica : $x_n = x_1 \times (r)^{n-1}$

Infinitésimo :

3. Para que f seja contínua $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x) = f(2)$

f é prolongável por continuidade a $x = 2$ pois $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$.

$(a \in \mathbb{R},\ b \in \mathbb{R}^+ \setminus \{1\}$ e $u = u(x),\ v = v(x))$

$a' = 0$	$(au)' = a.u'$
$(uv)' = u'.v + u.v'$	$\left(\frac{u}{v}\right)' = \frac{u'.v - u.v'}{v^2},\ v \neq 0$
$(u^a)' = a.u^{a-1}.u'$	$(b^u)' = u'.b^u.\ln(b)$
$(e^u)' = u'.e^u$	$(u^v)' = v.u^{v-1}.u' + v'.u^v.\ln(u)$
$(\ln(u))' = \frac{u'}{u}$	$(\log_b(u))' = \frac{u'}{u.\ln(b)}$
$(sen(u))' = u'.cos(u)$	$(cos(u))' = -u'.sen(u)$
$(tg(u))' = u'.sec^2(u)$	$(cotg(u))' = -u'.cosec^2(u)$
$(sec(u))' = u'.sec(u).tg(u)$	$(cosec(u))' = -u'.cosec(u).cotg(u)$
$(arcsen(u))' = \frac{u'}{\sqrt{1-u^2}}$	$(arccos(u))' = -\frac{u'}{\sqrt{1-u^2}}$
$(arctg(u))' = \frac{u'}{u^2+1}$	$(arccotg(u))' = -\frac{u'}{u^2+1}$

Pelo Teorema de Rolle, se $f(x)$ for contínua em $[a,b]$, diferenciável em (a,b) e $f(a) = f(b)$, então $f'(c) = 0$ existe.

Pelo Teorema de Lagrange, se $f(x)$ for contínua em $[a,b]$ e diferenciável em (a,b) então $f'(c) = \frac{f(b) - f(a)}{b - a}$.

4. Método de integração por partes $\Rightarrow \int u'.v\,dx = u.v - \int u.v'\,dx$

Método de integração por substituição $\Rightarrow \int f(x)\,dx = \int f(\varphi(t)).\varphi'(t)\,dx$

NÚMERO DE ALUNO(A): | | | | | | TURMA: | |

NOME: ______________________________

1. Considere a função $f : \mathbb{R} \to \mathbb{R}$ definida por

$$f(x) = 2\arctan(x-1) - x.$$

Determine os pontos de extremo e os pontos de inflexão de f.

$$f'(x) = 0 \Leftrightarrow 2 \times \frac{(x-1)'}{1+(x-1)^2} - 1 = 0 \Leftrightarrow \frac{2}{1+(x-1)^2} = 1 \Leftrightarrow 1+(x-1)^2 = 2 \Leftrightarrow (x-1)^2 = 1 \Leftrightarrow x-1 = \pm\sqrt{1} \Leftrightarrow$$

$$\Leftrightarrow x-1 = -1 \lor x-1 = 1 \Leftrightarrow x = 0 \lor x = 2$$

Pontos de extremo de $f(x)$: $x = 0 \lor x = 2$

$$f''(x) = \left(\frac{2}{1+(x-1)^2} + 1\right)' = \frac{\overset{0}{2'}.(1+(x-1)^2 - 2(1+(x-1)^2)'}{(1+(x-1)^2)^2} = -\frac{2 \times (2(x-1)'(x-1))}{(1+(x-1)^2)^2} = -\frac{4x-4}{(1+(x-1)^2)^2} = \frac{4-4x}{(1+(x-1)^2)^2}$$

$$f''(0) = \frac{4}{4} = 1$$

$$f''(2) = \frac{4-4.2}{(1+(2-1)^2)^2} = \frac{-4}{4} = -1$$

Pontos de inflexão de $f(x)$: $x = -1 \lor x = 1$

2. Calcule

$$\int 6x(1-x^2)^5\,dx.$$

$u = 1-x^2$ $\quad du = -2x$ $\quad dx = \frac{1}{du} \times du \Leftrightarrow$

$\Leftrightarrow dx = \frac{1}{-2x} \times du$

$$\int 6x(1-x^2)^5\,dx = 6\int x(1-x^2)^5\,dx = 6\int \cancel{x}(1-x^2)^5 . \frac{1}{-2\cancel{x}}\,du =$$

$$= 6\int (1-x^2)^5 . \frac{1}{-2}\,du = -3\int (1-x^2)^5\,du = -3\int u^5\,du = -3 \times \frac{u^6}{6} = -\frac{u^6}{2} = -\frac{(1-x^2)^6}{2} + C,\ C \in \mathbb{R}$$

Número de Aluno(a):

3. Calcule

$$\int (x^2-2)\ln(x)\,dx. \qquad \int u\,dv = uv - \int v\,du$$

$P(x^2) = \frac{x^3}{3} \quad (\ln(x))' = \frac{1}{x}$

$$\int (x^2-2)\times\ln(x)\,dx = \int x^2.\ln(x)\,dx - \int 2.\ln(x)\,dx = \frac{x^3}{3}\times\ln(x) - \int \frac{x^3}{3}\times\frac{1}{x}dx - 2\int \ln(x)\,dx =$$

$$= \frac{x^3.\ln(x)}{3} - \int \frac{x^{\cancel{3}}}{3\cancel{x}}\,dx - 2\int \ln(x)\,dx = \frac{x^3.\ln(x)}{3} - \frac{x^3}{9} - 2\int \underbrace{\ln(x)}_{u}\times\underbrace{1}_{dv}\,dx = \frac{x^3.\ln(x)}{3} - \frac{x^3}{9} - 2(\underbrace{\ln(x)}_{u}.\underbrace{x}_{v} - \int \underbrace{x}_{v}\cdot\underbrace{\frac{1}{x}dx}_{du}$$

$$= \frac{x^3.\ln(x)}{3} - \frac{x^3}{9} - 2\left(\ln(x).x - \int 1\,dx\right) = \frac{x^3.\ln(x)}{3} - \frac{x^3}{9} - 2\ln(x).x + 2x + C\,,\quad C\in\mathbb{R}$$

4. Calcule

$$\int \frac{x+3}{(x^2+3)(x-1)}\,dx$$

$$= \frac{Ax+B}{x^2+3} + \frac{C}{x-1}$$

$$\int \frac{x+3}{(x^2+3)(x-1)}\,dx = \int \frac{-x}{x^2+3} + \frac{1}{x-1}\,dx =$$

$$x+3 = (x-1)\times(Ax+B) + (x^2+3)C \iff$$

$$\iff x+3 = Ax^2 + Bx - Ax - B + Cx^2 + 3C$$

$$\iff \underset{1\quad 1\quad 3}{x+3} = (A+C)x^2 + (B-A)x + (-B+3C)$$

$$\begin{cases} A+C=0 \\ B-A=1 \\ -B+3C=3 \end{cases} \iff \begin{cases} A=-C \\ B+C=1 \\ -B+3C=3 \end{cases} \iff \begin{cases} A=-C \\ B=-C+1 \\ +C-1+3C=3 \end{cases} \iff$$

$$\iff \begin{cases} A=-C \\ B=-C+1 \\ 4C=4 \end{cases} \iff \begin{cases} A=-1 \\ B=0 \\ C=1 \end{cases}$$

$$= -\int \frac{x}{x^2+3}\,dx + \int \frac{1}{x-1}\,dx =$$

$$= -\int \frac{x}{x^2+3}\,dx + \ln(x-1) =$$

$t = x^2+3 \qquad t' = 2x \qquad dx = \frac{1}{dt}\,dt$

$$= -\int \frac{\cancel{x}}{x^2+3}\times\frac{1}{2\cancel{x}}\,dx + \ln(x-1) =$$

$$= -\frac{1}{2}\int \frac{1}{x^2+3}\,dx + \ln(x-1) =$$

$$= -\frac{1}{2}\int \frac{1}{t}\,dt + \ln(x-1) =$$

$$= -\frac{1}{2}.\ln(t) + \ln(x-1) =$$

$$= -\frac{1}{2}.\ln(x^2+3) + \ln(x-1) + C\,,\quad C\in\mathbb{R}$$

NÚMERO DE ALUNO(A):

5. Seja $f : [-2,2] \to \mathbb{R}$ uma função par. Sabendo que

$$\int_{-1}^{1}\left(\frac{3}{2} + f(x)\sin x\right)dx = \int_{0}^{1} f(x)\,dx \text{ e } \int_{2}^{1}(f(x)+\cos(\pi x))\,dx = 2$$

o valor de

$$\int_{-2}^{2} f(x)dx = 2\int_0^2 f(x)\,dx$$

??

é:

☐ -2 ☐ -1 ☐ $-\frac{\pi}{4}$ ☐ 0 ☐ 1 ☐ $\frac{\pi}{2}$ ☒ 2

6. A derivada da função $F : \mathbb{R}^+ \to \mathbb{R}$ dada por

$(\arctan(u))' = \frac{(u)'}{1-(u)^2}$

$$F(x) = \int_{-x}^{\frac{1}{x}} \arctan(t^2)dt.$$

$\arctan(1/x^2).(1/x^2) - \arctan((-x)^2).(-1) =$

$= \arctan(1/x^2).(-1/x^2) + \arctan(x^2)$

$= -\arctan(1/x^2).(1/x^2) + \arctan(x^2)$

é $\frac{d}{dx}\left(\int_{g(x)}^{f(x)} h(x)\,dx\right) = h(f(x)).f'(x) - h(g(x)).g'(x)$

☐ 0 ☐ $\arctan\left(\frac{1}{x^2}\right) - \arctan\left(x^2\right)$ ☒ $\arctan\left(x^2\right) - \frac{1}{x^2}\arctan\left(\frac{1}{x^2}\right)$

☐ $\frac{2x}{1+x^4}$ ☐ $\frac{1}{x}\arctan\left(\frac{1}{x^2}\right) + x\arctan\left(x^2\right)$ ☐ $-\frac{\frac{2}{x}}{1+\frac{1}{x^4}} + \frac{2x}{1+x^4}$

7. Fazendo a mudança de variável $x = 2\sin^2 t$ no integral

$x = 2\,\text{sen}^2 t$

$dx =$

$\frac{dx}{dt} = 2.\frac{d}{dt}(\text{sen}^2(t)) =$

$= 2.(2.\text{sen}\,t \times (\text{sen}\,t)') =$

$= 2.(2.\text{sen}\,t \times \cos t) =$

$= 4.\text{sen}\,t.\cos t$

$1 - \text{sen}^2 x = \cos^2 x$

$$\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{\sqrt{2(2-x)}}dx, = \frac{1}{\sqrt{2(2-2\,\text{sen}^2 t)}} = \frac{1}{\sqrt{4(1-\text{sen}^2 t)}} =$$

$= \frac{1}{\sqrt{4.\cos^2 t}} = \frac{1}{2.\cos t}$

obtemos o integral

$$\int_{A}^{B} f(t)dt,$$

com

	$\pi/6$	$\pi/4$	$\pi/3$
sen α	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$
cos α	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$
tan α	$\sqrt{3}/3$	1	$\sqrt{3}$

$A = \boxed{\pi/6}$, $B = \boxed{\pi/3}$, $f(t) = \boxed{2.\ \text{sen}\ t}$

$x = 1/2$

$1/2 = 2.\text{sen}^2 t \iff$

$\iff \text{sen}^2 t = 1/4$

$\iff \text{sen}\ t = 1/2$

$\iff t = \pi/6$

$x = 3/2$

$3/2 = 2.\text{sen}^2 t \iff$

$\iff 3/4 = \text{sen}^2 t$

$\iff \text{sen}\ t = \frac{\sqrt{3}}{2}$

$\iff t = \pi/3$

$f(t) = \frac{1}{2.\cos t} \times 4.\text{sen}\,t.\cos t =$

$= \frac{\cancel{4}.\ \text{sen}\ t.\ \cancel{\cos t}}{\cancel{2}.\cancel{\cos t}} = 2.\text{sen}\,t$